

on such devices is limited in its capacity. Thus it's onus on the operating system to make sure that the power is optimally utilized among all the hardware and applications.

## iOS

Albeit not the first smartphone OS to hit the market, it certainly is responsible for piquing everyone's fancy and opening the floodgates with its sleek and stylish UI. Originally presented as OS X for iPhone when it released in 2007, iOS is essentially a mobile version of OS X. It uses the ARM-variant of OS X's foundation – Darwin, which itself is built using XNU, a hybrid kernel which combines the Mach microkernel, elements of BSD and an object-oriented device driver API. It also shares a couple of application frameworks like Cocoa which powers the touch layer in iOS.

Applications for iOS are written in Objective-C. They interact with the operating system through abstraction layers – Core OS Layer, responsible for the low level features it's not made available to the developer directly but the other layers are built on top of this layer; Core Services Layer which provides all the fundamental system services that nearly all the applications use; Media Layer is responsible for handling graphics, video and audio subsystems and the Cocoa Touch Layer which is the front-facing component of the entire architecture responsible for managing multitasking, touch-based input, push notifications and other high-level system services. Apple provides developers with XCode and iOS SDK. The SDK exposes these layers to the



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developers which allow them to optimally access the system functionality. The current stable version of the operating system is 6.1.4. The next version which was previewed in WWDC 2013, iOS 7 is a major system overhaul which will be made available this fall.

iOS currently runs on iPhone, iPod Touch, iPad and Apple TV. Unlike the other systems that we'll discuss Apple doesn't allow the system to run on 3rd party hardware. This makes sure that Apple is in full control of the